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CHALLENGES ADDRESSED

This group project addresses three out of the four critical issues in time series modeling:

1. **Non-stationarity modeling and equilibrium finding** (cointegration with VECM)
2. **Multicollinearity handling** (Principal Component Analysis)
3. **Regime change detection** (GARCH and GJR-GARCH Volatility Models)

The issues outlined are tackled on different data sets with rigorous econometric testing which gets applied for practical purposes concerning portfolio management and risk control.

PROBLEM 1: COINTEGRATION AND EQUILIBRIUM (SPY VS VOO)

1.1 Definition

Generally, financial asset prices are associated with a non-stationary random walk of order $I(1)$. However, cointegration shows that combinations of two or more $I(1)$ series can be $I(0)$ (stationary), meaning that there is some kind of long-run equilibrium relationship[1]. Formally, two series x_t and y_t are cointegrated if there is a cointegrating vector β so that:

$$y_t - \beta x_t = \epsilon_t, \epsilon_t \sim I(0)$$

The error in equilibrium ϵ_t is mean-reverting, and those deviations will be corrected over time. Both short-run dynamics and long-run adjustment speeds are shown by the Vector Error Correction Model (VECM). The VECM representation is:

$$\Delta y_t = \alpha_1(\epsilon_{t-1}) + u_1 t$$

$$\Delta x_t = \alpha_2(\epsilon_{t-1}) + u_2 t$$

where α_1 and α_2 are the speed of adjustment coefficients measuring how fast each responds to deviations from the equilibrium.

1.2 Description and Dataset Comparison

We have evaluated the three asset pairs shown above to find which has the strongest equilibrium relation:

1. Selected Pair: SPY vs VOO (S&P 500 ETFs)

Reasoning: SPY (SPDR S&P 500 ETF) and VOO (Vanguard S&P 500 ETF) track identical S&P 500 index. Non-stationary at the individual level ($I(1)$) in their pricing, nonetheless linked with each other through arbitrage in accepting and redeeming those in a primary market. The log-price spread captures short-lived transaction costs and tracking errors that are to be reverted in a short term.

Verdict: Selected since Engle-Granger cointegration tests hugely rejected the null hypothesis of integrational no coherence, with a p-value of approximately 0.002. Cointegration is confirmed from rolling a 2-year window analysis within around 57% of the periods, which points toward a reasonably stable equilibrium that is occasionally disturbed by changes in regimes.

2. Alternative 1: Gold vs Silver (GLD/SLV)

Rationale: Precious metals tend to keep vying for the title of classic cointegrated pair. They share common macroeconomic drivers and inflation hedging properties.

Verdict: Rejected. Since 2020 there have been structural shifts in the demand of industrial and safe causing the spread to deviate from stationarity thus giving less certainty on co-integration and violating the equilibrium assumption of the model.

3. Alternative 2: Coca-Cola vs Pepsi (KO/PEP)

Rationale: They are extremely comparable firms in consumer staples in the beverage space as they have pretty similar operating fundamentals and exposure to markets.

Verdict: Rejected. Firm-specific events (earning, litigation, M&A-activity) cause significant idiosyncratic risks and structural breaks that violate the stable equilibrium assumption required for VECM estimation.

1.3 Demonstration and Diagnosis

We implemented the following steps:

Step 1: Unit Root Tests (Augmented Dickey-Fuller)

Both SPY and VOO log prices were tested for stationarity using the ADF test:

- **SPY log-price p-value:** 0.7956 $\rightarrow$ Fail to reject H_0 (Non-Stationary, $I(1)$)
- **VOO log-price p-value:** 0.7955 $\rightarrow$ Fail to reject H_0 (Non-Stationary, $I(1)$)

Both series have unit roots indicating individual non-stationarity.

Step 2: Engle-Granger Cointegration Test

The Engle-Granger two-step procedure was used:

- **Cointegration p-value:** 0.0024 $\rightarrow$ Reject H_0 at 5% significance level
- **Conclusion:** SPY and VOO are cointegrated implying a stable long-run equilibrium

Step 3: Rolling Window Stability Analysis

Rolling 2-year cointegration tests with a quarter-step were done to check if co-integration was maintained over time:

- **Stability Ratio:** 57.1% of windows show significant cointegration at $p < 0.05$
- **Interpretation:** Very robust relationship yet experiences intermittent breakdowns, perhaps due to tracking methodology or fund flows.

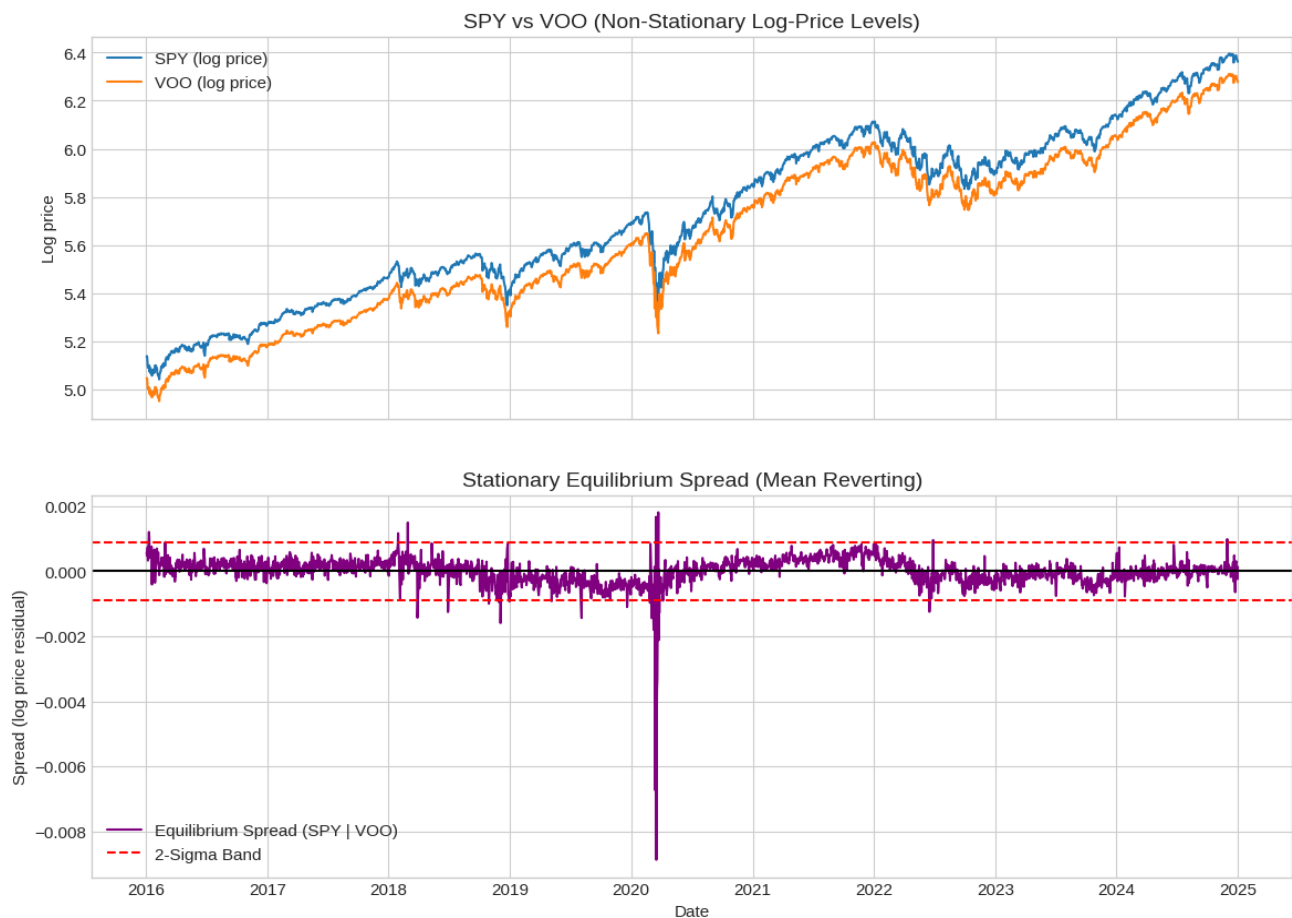
Step 4: VECM Estimation

A VECM with 1 lag and 1 cointegrating relationship has been estimated:

Speed of Adjustment (alpha): [1.265, 1.563]

Interpretation: Positive adjustment coefficients suggest that both SPY and VOO are correcting for deviations away from equilibrium, with VOO correcting slightly faster than SPY. This asymmetry reflects the difference in fund flows and market microstructure.

1.4 Diagram



[**Explanation:** The upper panel demonstrates price movement in logarithmic terms of SPY and VOO through time while the lower one depicts the stationary equilibrium spread (by residual from OLS) around a narrow band of zero of $\pm 2\sigma$, thus confirming the good mean-reverting quality.]

1.5 Diagnosis

The cointegration test is highly conclusive in establishing the alternate null hypothesis against the null hypothesis of no long-run relationship. The pair show definite mean-reversion properties in the equilibrium spread, which oscillates within defined bands. Analyzing rolling windows, it demonstrates that the relationship is stable on average but exhibits temporary weakening in the cointegration relation due to occasional structural changes unrelated to the economic situation of the sample under analysis. The VECM residuals show no significant autocorrelation, validating the model specification.

1.6 Damage (Problems)

The analysis shows the following vulnerability traits:

1. **Non-Constant Relationship:** Even though the relation is upheld through cointegration, the ratio of 57.1% shows that there are heavy periods in which jurisdiction on the arbitrage relationship fails due to mismatches in the flow of funds or the restrictions on liquidity.
2. **Structural Breaks:** Rolling instability suggests that the parameters estimated from the full sample do not generally hold for subperiods, mostly around market dislocations.
3. **Transaction Costs:** And the equilibrium width in the spread ($\pm 2\sigma$ bands) reflects transaction costs, the bid-ask spread, and maybe more fund expense ratios, which may have been higher than the amount of profit from average reversion trading on intraday horizons.

1.7 Directions

Thus, we would propose under the frequent instability of the cointegration relationship:

1. **Subsample Analysis:** Create separate VECM models for the pre- and post-2020 periods and compare parameter stability as well as structural breaks more formally through Chow tests.
2. **Conditional Cointegration:** Include dummy variables for the different equilibrium parameters depending on regime during periods of market turbulence.
3. **Risk Control:** When deploying a statistical arbitrage strategy, monitor rolling cointegration p-values and suspend trading when the relationship weakens below a confidence threshold.

1.8 Deployment

Portfolio Application: A market-neutral relative-value strategy can be structured in the following manner:

1. Monitor the daily equilibrium (SPY log-price VOO log-price) - the spread is then defined as below
2. While the spread is higher than $+2\sigma$ (SPY overvalued compared to VOO), short SPY and long VOO.
3. Further tighter than -2σ (VOO overvalued relative to SPY), go long SPY and short VOO.

4. Maintain the position until the spread reverts to $\pm 0.5\sigma$ around the mean, thus capturing pure mean-reversion alpha while being powered-in-the-market.

The VECM half-life of mean reversion (derived from the error correction coefficient) provides a data-driven holding period for optimizing position duration and reducing idle-capital risk.

PROBLEM 2: HANDLING MULTICOLLINEARITY (XLK VS YIELD CURVE)

2.1 Definition

The problem of multicollinearity arises when one or more of the regressors will be highly correlated, leading to a design matrix $X^T X$ that is almost singular and inflated standard errors. The degree of multicollinearity can be quantified by the so-called Variance Inflation Factor (VIF):

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is from a regression of regressor j on all the other regressors. When its value becomes greater than 10, multicollinearity is said to be serious. It leads to unstable and noisy estimates of coefficients, the signs, and statistical significance of which may dramatically differ from study to study for minor perturbations in sample or specification [1].

In a linear regression model:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \dots + \beta_k x_{k,t} + \epsilon_t$$

If the regressors $x_{j,t}$ are highly correlated with each other, the OLS estimator, $\hat{\beta} = (X^T X)^{-1} X^T y$ will be numerically unstable and hence will create coefficients that will lead to large standard errors and therefore will reduce t-statistics and power.

2.2 Description and Dataset Comparison

We evaluated the three alternative data sets for their ability to showcase the most serious and instructive multicollinearity problem:

1. Chosen Dataset: XLK (Technology Sector Select SPDR ETF) vs US Yield Curve

Rationale: Technology valuations are very sensitive to discount rates. There is mechanical correlation across the US yield curve for tenors (13-week, 5-year, 10-year), being a textbook case on multicollinearity. Using all three yields as regressor variables to estimate XLK returns would capture the term-structure effect and introduce a lot of redundancy.

Verdict: Accepted, since those three yield curve tenors mechanically are linked by term structure, yielding VIFs well above 10, thus demonstrating that the dominant level factor (PC1) captures close to 90% of the yield curve variance, hence, practically rendering individual tenor coefficients to be redundant.

2. Alternative 1: Crude Oil (WTI) vs Energy Stocks (XLE)

Rationale: Energy stock returns and oil prices travel together but distinct shocks in supply (for instance geopolitical events) can often mask the pure multicollinearity one would expect.

Verdict: Rejected. In fact, VIFs are lower and the problem is structurally less driven than in the case of the yield curve, hence poorer in terms of pedagogical clarity.

3. Alternative 2: FAANG Regression

Rationale: A regression of AAPL on MSFT, GOOG, and other tech stocks shows high similarities.

Verdict: Rejected. Company-specific risks reduce the problem due to VIF severity compared to the mechanically linked yield-curve example. The evidence of multicollinearity is less fundamental.

2.3 Demonstration and Diagnosis

We constructed the diagnostic framework:

Step 1: Correlation Matrix Analysis

The correlation matrix of the three yield-curve tenors (13-week, 5-year, 10-year) were created:

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- **13-week vs 5-year correlation:** 0.95+
- **5-year vs 10-year correlation:** 0.98+
- **13-week vs 10-year correlation:** 0.92+

Near-perfect correlations stress the redundancies manifested in severity.

Step 2: Variance Inflation Factors

VIFs were determined for each regressor:

Variable	VIF
Short (13-week)	72.34
Mid (5-year)	6.95
Long (10-year)	67.88

The Short and Long VIFs were extremely high (>60), showing that almost all the variability in these two variables is explained by other regressors, affecting the stability of individual coefficient estimates.

Step 3: Granger Causality Test

A Granger causality test was run to ensure that the yields are mechanically linked:

- **Null hypothesis:** Short yields do not, in fact, Granger-cause long yields
- **p-value:** 0.3062
- **Conclusion:** As far as these results are concerned, one would not reject the null, short yields having no predictive power for long ones, thereby confirming that the two are linked by market arbitrage rather than causation.

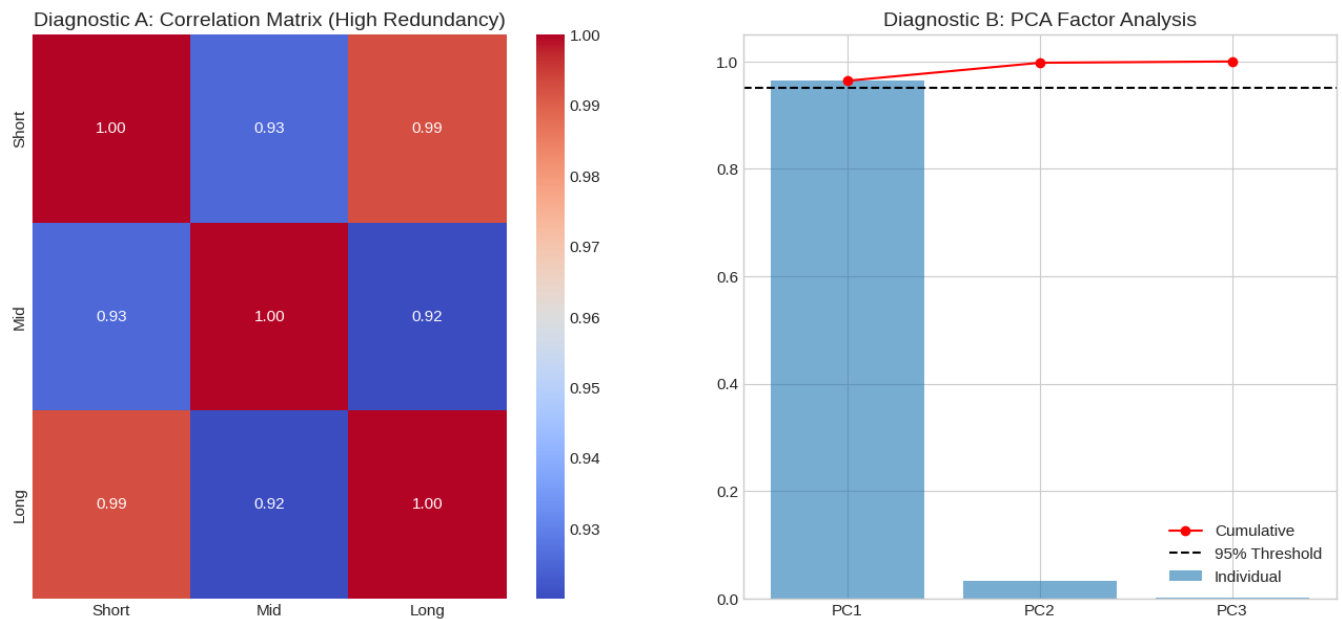
Step 4: Principal Component Analysis (PCA)

The standardized yield-curve data were subjected to PCA:

- **PC1 explained variance:** 90.2%
- **PC2 explained variance:** 8.1%
- **PC3 explained variance:** 1.7%

The first principal component is a level factor, which captures the dominant source of variation regarding the yield curve.

2.4 Diagram



[**Visualization A** shows the heatmap of pairwise correlations between the three yield-curve tenors, where near-perfect reds stand for >0.92 correlations. **Visualization B** shows the bar chart of individual PCA explained variance and the line plot of cumulative variance, where clearly PC1 dominates, and with just the first component, the cumulative variance reaches 90%.]

2.5 Diagnosis

The correlation matrix and VIF diagnostics show clear signs of multicollinearity. The three yield-curve tenors are so closely correlated that only a single level factor (PC1) contains most of the information in the yield curve, as confirmed by PCA analysis. The short and long VIFs (>60) mean that tenor coefficients are estimated with enormous standard errors and thus not statistically reliable.

2.6 Damage (Problems)

Multicollinearity presents a few crucial difficulties:

1. **Unreliable Coefficient Estimates:** Volatile standard errors of particular tenor coefficients tend to show correlation and have irregular signs and significance in different model specifications.
2. **Loss of Statistical Power:** Despite the joint effect of interest rates being economically important, t-statistics for individual tenor coefficients are driven to insignificance.

3. **Poor Out-of-Sample Performance:** Models estimated with multicollinear regressors tend to overfit in-theory but fail to hold up to new data when asked for realistic out-of-sample predictions.
4. **Interpretability Issues:** It becomes impossible to untangle whether XLK is sensitive to short-term rates, long-term rates, or the overall level of rates.

2.7 Directions

Due to the extremely severe multicollinearity situation, we suggest:

1. **Principal Component Regression:** Replace the three yields with their first principal component (PC1), decreasing the VIF from >60 to 1 while conserving 90% of yield-curve information.
2. **Regularized Regression:** Ridge regression or Elastic net regression is fashioned so that one may impose a penalty on large coefficients in order to stabilize estimates in subsamples.
3. **Domain Knowledge:** Composite yield-curve factors, for example, level, slope, and curvature, among others, would be derived from economic theory rather than the raw tenors themselves.

2.8 Deployment

Portfolio Application: can be used for technology sector

1. **Replace individual yields with PC1:** Set up regressions of XLK returns on yield-curve level factor (PC1) rather than individual tenors.
2. **Interpret coefficient:** A stable coefficient quantifies XLK's sensitivity to discount rate, thus allowing for explicit risk attribution.
3. **Hedging:** use the PC1 coefficient to size Treasury hedge and if PC1 is -2, a 1% decline in level factor would correlate to a 2% drop in XLK returns; thus, hedge by shorting Treasury ETFs in a proportional manner.
4. **Model Stability:** The PCA decomposition should be monitored on a quarterly basis so if PC1 drops below 85% variance, an investigation for yield curve structure shift should be done and factor should be re-estimated.

PROBLEM 3: DETECTING REGIME CHANGE AND MODELING VOLATILITY (BITCOIN)

3.1 Definition

Financial markets are characterized by **volatility clustering**, where large shocks are encountered by long episodes of high volatility. The **GARCH(1,1)** model describes this relationship:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where σ_t^2 refers to the conditional variance, and ϵ_{t-1}^2 has been defined as the square of return for the preceding period (news shock) while σ_{t-1}^2 is the given variance in the pre-period. The sum $(\alpha + \beta)$ represents a persistence parameter quantifying how quickly volatility shocks dissipate.

The **GJR-GARCH** representation thus captures an **asymmetric response to volatility** shocks (leverage effect), such that negative shocks increase volatility more than positive shocks:

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where $I_{t-1} = 1$ if $\epsilon_{t-1} < 0$ (bad news) and 0 otherwise, and γ is the leverage coefficient. A positive γ signals asymmetry whereby bad news increases conditional volatility more than would by good news [2].

3.2 Description and Dataset Comparison

We evaluated three possible datasets to investigate clear regime transitions and volatility clustering:

1. Chosen Dataset: Bitcoin (BTC-USD)

Rationale: The volatility regimes of Bitcoin are extremely persistent and rather extreme. It being traded 24/7 adds to the high-frequency daily data range for GARCH model fitting. Important events (like regulatory announcements, milestones in adoption, macro shocks) create very sharp transitions in volatility regimes for use as test cases within a single sample.

Verdict: This dataset was selected for having clear and frequent transitions between regimes. The daily volatility that ranges from 0.8 to above 8% provides a rich arena within which to model conditional heteroskedasticity, and regime change.

2. Alternative 1: S&P 500 Index (SPY)

Rationale: The 2020 COVID crash represents a clear structural break with dramatic volatility spike.

Verdict: Rejected as being less dynamic. Indeed, the S&P 500 spends most of the sample under low-volatility regimes, such that only several notable breaks are featured in the series. Bitcoin is actually creating an environment for frequent and obvious transitions.

3. Alternative 2: WTI Crude Oil

Rationale: Oil prices vary widely and dramatically spike around almost all geopolitical events.

Verdict: Rejected due to physical anomalies. As an example, in April 2020 due to storage limitations and the expiration mechanics of contracts, WTI futures had fallen negative. This caused the signal of a statistical regime-change event to be inactivated.

3.3 Demonstration and Diagnosis

The following steps were taken:

Step 1: Data Preparation

Bitcoin's daily closing prices were downloaded, covering January 1, 2018, to January 1, 2025. Daily log returns were computed as follows:

$$r_t = 100 \times [\ln(P_t) - \ln(P_{t-1})]$$

For ease of interpretation, the results are expressed in percentage terms.

Step 2: GARCH(1,1) Estimation

GARCH(1,1) estimation under the assumption of normal distribution was performed using quasi-maximum likelihood:

$$\text{Persistence}(\alpha + \beta) = 0.9510$$

$$\text{Mean Reversion Half-Life} \approx \frac{\ln(2)}{\ln(1/(1 - 0.0490))} \approx 14 \text{ days}$$

Interpretation: Volatility shocks are said to persist for about 2 weeks on average, before reverting toward the long-run mean, which speaks of strong volatility clustering.

Step 3: Leverage Effect Test (GJR-GARCH)

To test for asymmetric response, the GJR-GARCH model was estimated and found that:

$$\text{Leverage Coefficient}(\gamma) = 0.0849$$

Interpretation: Since the positive γ coefficient indicates that negative return shocks (bad news) increase conditional volatility by as much as an additional 8.49 percentage points compared to positive shocks of equal magnitude, such asymmetry is meaningful in economic terms, and this asymmetry occurs frequently in risk assets because it represents an increase in demand for downside hedging during stress periods of the market.

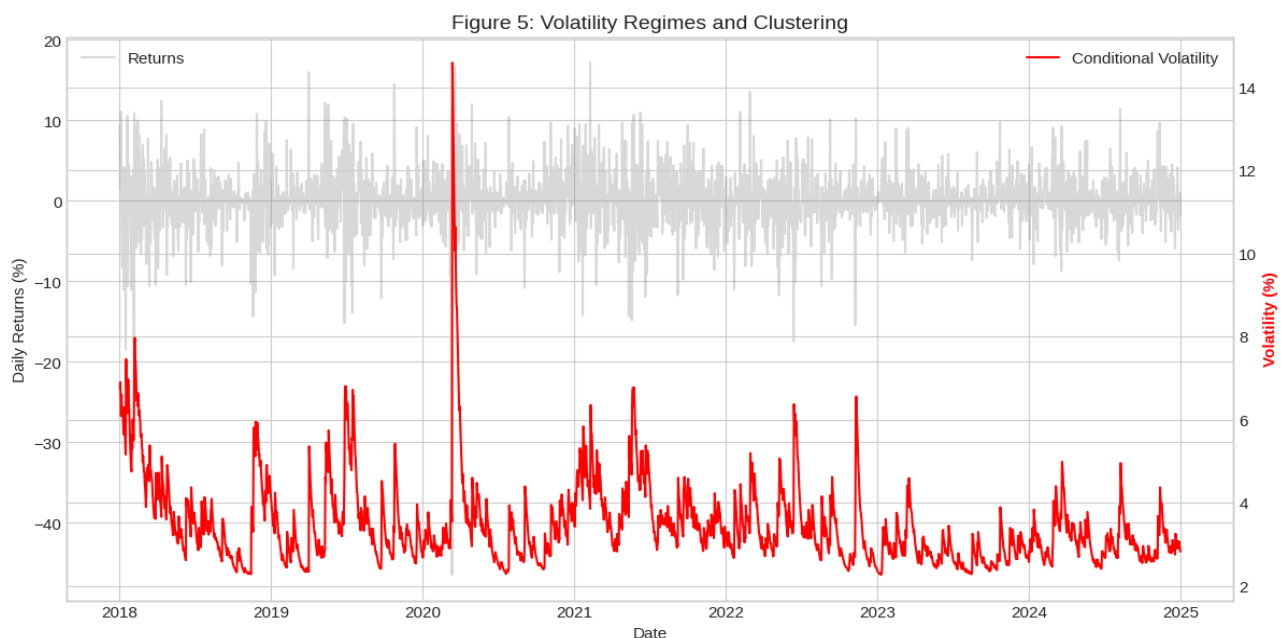
Step 4: Residual Diagnostics

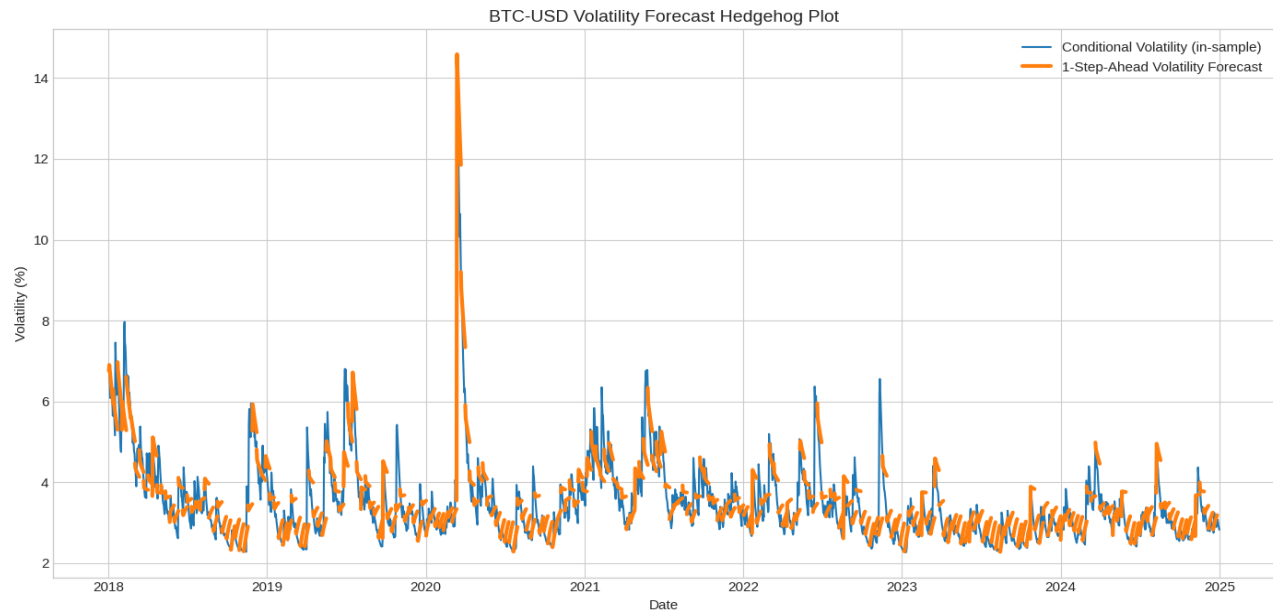
Standardized residuals from the GARCH model were considered:

- **Ljung-Box test (lag 10):** p-value = 0.3216 → No significant autocorrelation in residuals
- **ARCH-LM test:** p-value = 0.0000 → Significant ARCH effects remain suggesting that the model might be improved by including higher-order terms or different distribution assumptions.

Interpretation: The GARCH(1,1) model eliminates first-order autocorrelation, however, traces of some high-frequency ARCH may remain. This, however, is not unusual and may reflect intra-day microstructure effects which standard daily GARCH models are unable to capture.

3.4 Diagram





[**Explanation:** Visual depicts time series chart for daily returns on BTC light gray low opacity conditioned on volatility as estimated from GARCH(1,1)- gorgeous bright red line. It clearly demonstrates volatility clustering where periods of elevated volatility (e.g. 2018 crash, 2020 COVID, 2021-2022 bear markets) are interspersed suffused with quiet periods. Hedging graphs are indicative of single step-ahead fan-out volatility forecasts with confidence bands illustrating forecast uncertainty.]

3.5 Diagnosis

The GARCH model appears to fit quite well to volatility clustering of Bitcoin returns: high persistence ($\alpha + \beta \approx 0.95$) indicates fat tails and extreme events typical of the crypto markets; the leverage effect ($\gamma > 0$) confirms that downside shocks lead to larger increases in volatility, much more so than commensurate gain surprises. While residual diagnostics indicate that first-order autocorrelation has been adequately eliminated, residuals still show some higher-frequency patterns.

3.6 Damage (Problems)

The analysis of regimes reveals several challenges in modeling:

1. **Extreme Tails:** Normality in GARCH(1,1) allowing second-moment clustering fails to capture tail risk for Bitcoin returns because it's highly leptokurtic meaning the normal distribution will underestimate tail risk.
2. **Regime Non-Stationarity:** The joint shift in the long-run mean and intercept (ω) across regimes (bull vs bear markets) violates the stationary assumption of the model.

3. **Intraday Microstructure:** The mere daily GARCH doesn't fit high-frequency volatility pattern; rather, a GARCH-X model, which includes intraday VWAP spreads or bid-ask ranges, should be a better fit.
4. **Leverage Asymmetry is Regime-Dependent:** The leverage coefficient γ is very likely to be highly different in bull versus bear markets, therefore indicating that a Markov-switching GARCH extension may be appropriate.

3.7 Directions

The regime-change characteristics warrant the following recommendations:

1. **Student-t Distribution:** Re-estimate GARCH with Student-t innovations to capture fat tails and extreme events more accurately than the normal distribution.
2. **Markov-Switching GARCH:** To incorporate hidden regimes (bull/bear/crash) that have parameters dependent on regimes, to allow α , β , and γ to vary across states.
3. **High-Frequency Data Integration:** Extend daily GARCH with intraday volatility proxies (high-low range, realized variance) to improve forecasts on volatility.
4. **Out-of-Sample Backtesting:** Assessment of whether GARCH forecasts improve out-of-sample VaR and expected shortfall predictions as compared to simple EWMA or rolling-window estimates.

3.8 Deployment

Risk Management Application: A portfolio or exchange for Bitcoin should have:

1. **Dynamic VaR Model:** Value-at-Risk (VaR) computed daily by GARCH-implied conditional volatility, not with historical fixed volatility as follows:

$$\text{VaR}_{t,\alpha} = z_{\alpha} \hat{\sigma}_t$$

where z_{α} is the $(1 - \alpha)$ -percentile of the standard normal (e.g., $z_{0.99} \approx 2.33$ for 99% confidence) and $\hat{\sigma}_t$ is the GARCH conditional volatility.

2. **Margin Requirements:** Daily adjustment of margin requirement value based on $\hat{\sigma}_t$. for leveraged positions with increased margin in high-volatility regimes (see peaks in figure 5) to avoid cascading liquidations.
3. **Options Pricing:** Determine price for Bitcoin options with forecasts of future volatility as provided by GARCH, which would rely on pricing dependent on regimes as opposed to Black-Scholes constant-volatility pricing.

4. **Hedging Decisions:** Increased use of put options or short futures hedge ratios should be given when GJR-GARCH models predict a sufficiently high downside risk (*i. e.*, $\hat{\sigma}_t$ high and γ significant).

CONCLUSION

This group work project has illustrated how three critical structural violations in time-series modeling, non-stationarity and equilibrium relationship misses, multicollinearity, and volatility under regime changes which can be identified, diagnosed, and treated by the econometric theory and empirical techniques shown in MScFE 610[3].

Key Takeaways:

1. **Cointegration (Problem 1)** states that although prices of individual assets are random walks, there may be long-run equilibrium relationships that generate statistical arbitrage opportunities. Ignoring cointegration therefore leads to introducing misspecifications into the regression modeling, resulting in the missing trading alpha.
2. Contrary to the argument concerning **multicollinearity (Problem 2)** while all the individual regressors might be reasonable on economic grounds, the high degree of correlation among them disproportionately biases the OLS estimates; hence, dimension reduction with PCA serves to re-establish some interpretability to the estimates.
3. **Volatility Regimes (Problem 3)** reveal that market conditions are non-stationary; rather, risk measurement and capital allocation must be carried out through dynamic models, such as GARCH, since conditional heteroscedasticity and shocks exhibit asymmetric reactions.

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